

Individual Portfolio and Contribution Record

STUDENT	Rikesh Aryal
STUDENT ID	cihe251435
GROUP	5
PROJECT TITLE	Healthcare integration test harness
ROLE IN GROUP	Research & literature lead
SECTIONS OWNED	make SNOMED CT-AU coding structure, address ethics and draft patient data model
GITHUB REPOSITORY	https://github.com/SkinnySeal17/HEAALTHCARE_SYSTEM
PROJECT BOARD	https://student-team-siuckpgv.atlassian.net/jira/software/projects/HITH/list?jql=project%20%3D%20HITH%20ORDER%20BY%20cf%5B10019%5D%20ASC
GENERATED	13 Sept, 13:45

ACTIVITY SUMMARY — GENERATED FROM THE LOG, NOT SELF-REPORTED

MEASURE	WEEK 6	WEEK 7	WEEK 8	TOTAL
Entries logged	1/5	6/6	///	16/20
Minutes on task	20	5	8	16
Evidence items	7	4	9	14
Completeness	100%	100%	100%	80%

Time recorded during scheduled class: 1 minutes. Time recorded outside class: 14 minutes.

1 · Individual responsibility summary

ASSESSMENT SECTION	INDIVIDUAL RESPONSIBILITY	EXPECTED OUTPUT	TARGET
4. solution design 4.1 and 4.2	dsign solution architecture and tech stack justification	a finalised teck stack and solution architecture of test harness	week 6&7
4.3 and 4.4	finish 4.1 and 4.2 and roughly sketch security and cost estimation table	a rough security flow and cost estimation to be reviewed	week 7&8
4.4	write out the cost estimation and finalise it	a finalised security flow and cost analysis with tech used	Week 8

Week 6

5 of 5 entries logged · 20 minutes recorded · 7 evidence items · 100% complete

W6-1 · Frame the problem in your own words

Logged · 5 min · 3/3 required parts · 31 Aug, 09:06

The problem, as you understand it

MedTalk Connect (the real, existing interoperability platform) sits between MedTalk AI and four clinic systems (Best Practice/Halo Connect, Oracle Health Millennium, Epic, MedicalDirector), translating canonical FHIR R4 data into vendor-specific formats. There is currently no safe, repeatable, data-rich environment to test that this pipeline — and

s p e c i f i c a l l y t h e c l i n i c a l c o d i n g l a y e r (S N
degraded network conditions, without touching real patient data or real production systems..

b u i l d a s y n t h e t i c F H I R R 4 s a n d b o x w i t h m
layer, and an automated test harness that produces a structured pass/fail report — deployed to a real testing environment
for live demonstration

One thing that is deliberately OUT of scope, and why

he project is explicitly focused on system interoperability rather than user interface capabilities. Specifically, the technical scope is limited to enhancing the underlying core AI transcription and natural language processing engine behind MedTalk. Therefore, modifying how the scribe tool itself functions or operates for clinicians will not be addressed, because the technical team is exclusively focusing on backend integration updates. This clear distinction ensures the development remains dedicated solely to structural connectivity instead of adding new front-facing features right now.

Assessment 2 section you worked on today

SOLution architecture

W6-1-01 · 31 AUG 2026

- Implementation or coding of the test harness (extended to Stage 2).
- Real patient data or access to real production EMR.
- This project is about interoperability and not about the functionality of the scribe, it's about changes to the core AI transcription/NLP engine of MedTalk.

Section: — · Task: Frame the problem in your own words

W6-1-02 · 31 AUG 2026

- Implementation or coding of the test harness (extended to Stage 2).
- Real patient data or access to real production EMR.
- This project is about interoperability and not about the functionality of the scribe, it's about changes to the core AI transcription/NLP engine of MedTalk.

Section: — · Task: Frame the problem in your own words

W6-2 · Literature search log

Logged · 1 min · 6/6 required parts · 31 Aug, 09:52

Search strings you actually typed

FHIT data, sythetic patient data AU core guide

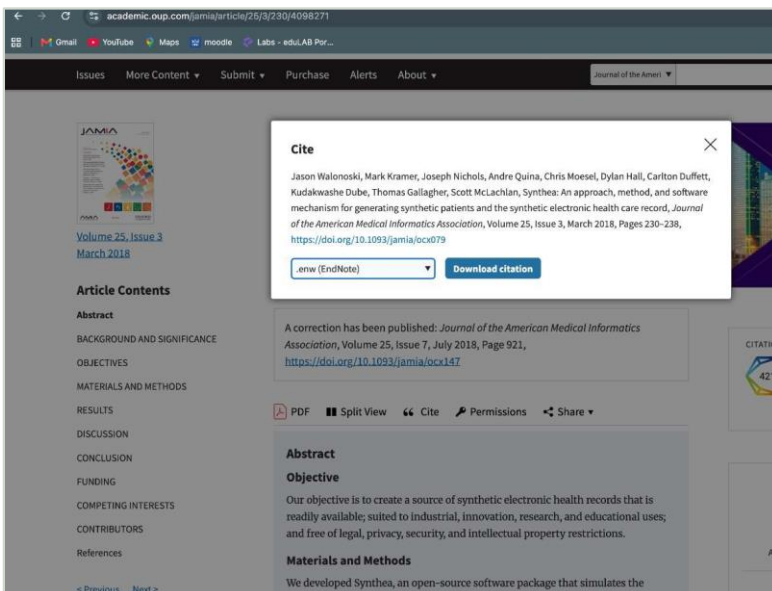
Databases / sites searched

sciencedirect, national library of medicine , google

SOURCE (AUTHOR, YEAR, TITLE)	LINK OR DOI	KEY INFORMATION YOU OBTAIN	HOW IT WILL BE USED
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SOURCE (AUTHOR, YEAR, TITL	LINK OR DOI	KEY INFORMATION YOU OBTAIN	HOW IT WILL BE USED
Walonoski, J., Kramer, M., Nichols, J.L., Quina, A., Moesel, C., Hall, D., Duffett, C., Dube, K., Gallagher, T. & McLachlan, S.	https://doi.org/10.1093/jamia/ocx079	SNOMED and HL7 structure record	it will supports our decision to generate a large synthetic dataset
Gruendner, J., Gulden, C., Kampf, M., Mate, S., Prokosch, H.-U. & Zierk, J.	http://dx.doi.org/10.2196/25645	how FHIR data can be queried directly with a preprocessing service and be used for statistical analyses.	PostgreSQL + JSONB for FHIR storage (justifies our database choice
2021			
AU Core Implementation Guide v1.0.0	https://hl7.org.au/fhir/core/1.0.0/index.html	AU Core defines the data model and RESTful FHIR API interactions that set minimum expectations for a system to record, update, search, and retrieve core digital health and administrative information, and provides the basis for further implementation guide development for specific use cases	justifies our profile references

W6-2-01 · 31 AUG 2026



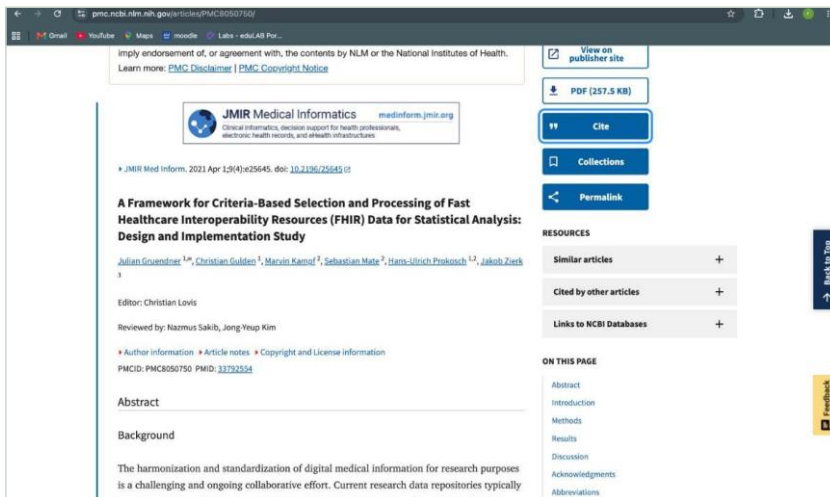
Section: proposed solution · Task: Literature search log

My contribution: search and see patient sample log to propose stack.

What it demonstrates: one million synthetic patient records are freely available online, encoded in standard formats including HL7 FHIR and Consolidated-Clinical Document Architecture, and accessible through an HL7 FHIR application programming interface.

<https://doi.org/10.1093/jamia/ocx079>

W6-2-02 · 31 AUG 2026



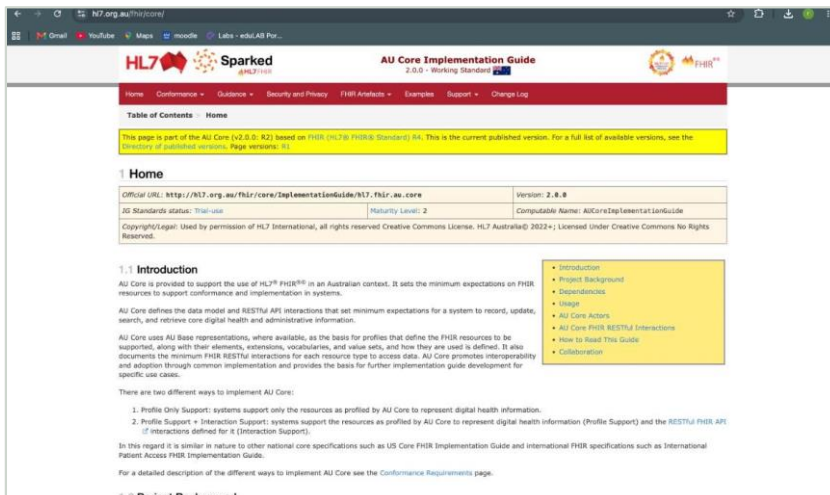
Section: tech stack justification · Task: Literature search log

My contribution: justify the use of postgres

What it demonstrates: how FHIR data can be queried directly with a preprocessing service and be used for statistical analyses.

<https://doi.org/10.2196/25645>

W6-2-03 · 31 AUG 2026



Section: 4.1 & 4.2 · Task: Literature search log

My contribution: HL7 core research

What it demonstrates: AU Core defines the data model and RESTful FHIR API interactions

<https://hl7.org.au/fhir/core/>

W6-3 · Claim your tasks on the board

Logged · 4 min · 4/4 required parts · 31 Aug, 09:14

Tasks you created or took ownership of (one per line, with task ID)

HITH-35 proposed solution architecture (due EOM)

HITH-36 justification of stack (EOM)

Your contribution to the charter, scope, requirements, RACI, risk register or feasibility study

Scope: Defines the technical boundaries. Your architecture explicitly outlines what systems, APIs, and data flows are In-Scope versus legacy components left Out-of-Scope. Feasibility Study: Proves viability. The stack justification validates that your chosen technologies can realistically support the performance, compliance, and scaling needs within budget. Risk Register: Uncovers technical threats. Your design identifies architectural single points of failure, integration bottlenecks, or open-source dependencies to mitigate. Requirements: Grounding Non-Functional Requirements (NFRs). The stack justification serves as the concrete proof that your system will hit mandatory security, uptime, and latency

benchmarks.Project Charter: Visualizes the vision. Your high-level architecture provides the technical blueprint that reassures stakeholders how the business goals will be structurally achieved.RACI Matrix: Establishes ownership. You are Responsible (R) or Accountable (A) for these technical designs, while mapping out who must be Consulted (C) (e.g., security, infrastructure teams).

W6-3-01 · 31 AUG 2026

Q Search backlog	BA C SP	Filter		
☒	HITH-34	Technology and Market Survey	☐ In Progress	C
☒	HITH-35	Proposed Solution Architecture	☐ In Progress	RA
☒	HITH-36	Technology Stack and Justification	☐ In Progress	RA
☒	HITH-37	Sprint Plan	☐ In Progress	
☒	HITH-45	Proof of Concept Description	☐ In Progress	

Section: 4.1 &4.2 · Task: Claim your tasks on the board

My contribution: i proposed tech and solution based on what we are building

What it demonstrates: task division and im doing sonmething, RA is me with work in progress

<https://student-team-siuckpgv.atlassian.net/jira/software/projects/HITH/boards/34/backlog?selectedIssue=HITH-36>

W6-4 · Draw the first figure

Logged · 3 min · 5/5 required parts · 31 Aug, 09:26

What you drew

As-Is process flow

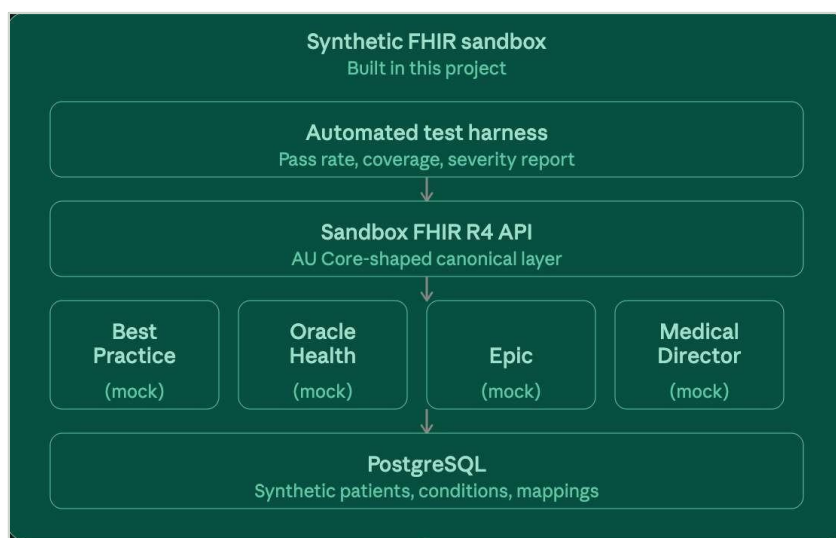
Explain the figure — what a reader should take from it

the teal container is everything we're actually building — test harness, sandbox FHIR API, the four mock connectors side by side, and PostgreSQL underneath which will in SYNC using API will deduce a result to check the working of final product. they are divided in 4 sections data sandbox api, mock connector and test harness and reporting layer that will give result.

One part of the figure you are not confident about

im confident about the overall flow but some might be changed during the actual production phases like mockup provided by FAZ and also the medtalk connector synthesis.

W6-4-01 · 31 AUG 2026



Section: proposed solution · Task: Draw the first figure

My contribution: this figure is based on the file provided by fazaus on what we are building . i have included it in my pr solution stack.

What it demonstrates: It represent the overall flow of test harness.

https://github.com/SkinnySeal17/HEALTHCARE_SYSTEM/tree/main/DIAGRAMS

W6-5 · Weekly reflection

Logged · 8 min · 4/4 required parts · 31 Aug, 09:33

What did I personally complete this week?

In week 6, i wrote a comprehensive section about the proposed solution relating to FHIR sandbox making along with the tech stacks that will be used to make and manage it. The justification is also written to show the mentor as he has also proposed a tech stack and make a file for review.

What did I learn about the problem or the technology?

Large-scale implementation studies (such as at University Hospital Erlangen-Nürnberg) successfully managed over 35.5 million FHIR resources using PostgreSQL's binary JSON format. This proves that PostgreSQL + JSONB is an

industry-proven, highly scalable solution for semi-structured health data, eliminating the need for complex, dedicated document stores. My research shows that handling the variations makes absolute server validation a heavy, complex task. This justified narrowing our project scope to focus strictly on structural validation via lightweight JSON Schema checks, rather than deploying heavy, official HL7 validator tools

What decision did I influence, and why?

Stack Dictation: my work on HITH-36 (Justification of Stack) locked in PostgreSQL, Docker Compose, and free-tier hosting, forcing the architecture to fit these exact technical boundaries.

Scope Constraints: By ruling full AU Core conformance out of scope, I steered the architectural design toward a lightweight JSON Schema validation approach.

Security & Cost Ownership: Because I am Responsible and Accountable (R/A) for Security Design & Cost (4.3–4.4), the architecture had to conform to My rules around free-tier deployment limits and sandbox isolation - also to be confirmed and reviewed .

What will I finish before the next class?

Before next class, I will complete the tech stack selection and technical justification (HITH-36) alongside a rough draft of the proposed solution architecture (HITH-35) for immediate peer review. Once feedback is gathered from the team, I will finalize the official architecture and stack documentation in the coming week.

Week 7

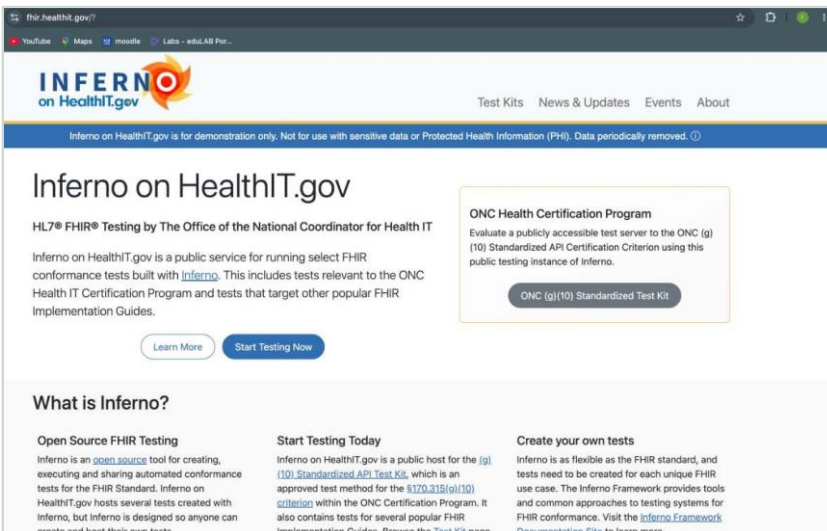
6 of 6 entries logged · 5 minutes recorded · 4 evidence items · 100% complete

W7-1 · Compare three existing tools

Logged late · 1 min · 4/4 required parts · 07 Sept, 08:37

TOOL / SYSTEM / APPROVCEHNDOR & LICENCE		FEATURES RELEVANT TO OUR P	LIMITATION WE CARE ABOUT	SOURCE
Synthea	MITRE Corporation / Open source, Apache 2.0	Generates realistic but synthetic patient populations and clinical histories. Supports FHIR R4, including conditions, medications, observations, procedures and other clinical resources. Useful for producing safe test data without real patient information.	Primarily focuses on synthetic patient/data generation. It does not by itself provide the complete integration-testing workflow required for FHIR exchange, fault injection, ICD-10-AM/MBS mapping validation and automated test reporting.	Github
HAPI FHIR	Smile CDR / HAPI FHIR open-source project	Provides a FHIR implementation and server framework that can be used to store and exchange FHIR resources. Useful for creating a FHIR-based test environment and testing API interactions.	Provides the FHIR infrastructure, but the project still needs additional testing logic for the specific requirements of this project, such as fault injection, mapping validation, automated pass/fail evaluation and project-specific reporting.	https://hapifhir.github.io/hapi-hl7v2/license.html?
Inferno	MITRE / ONC ecosystem, open source, Apache 2.0	Automated FHIR conformance testing framework. Provides executable test kits for FHIR implementation guides and can test FHIR APIs against defined requirements. It also provides simulated FHIR APIs for some test kits.	Strongly focused on FHIR conformance testing and specific implementation guides. It does not directly provide the complete combination of synthetic clinical data generation, SNOMED CT-AU ICD-10-AM / MBS mapping checks, healthcare-specific fault scenarios and the customised test harness required by this project.	https://fhir.healthit.gov/test-kits/

W7-1-01 · 07 SEPT 2026



Section: AS_is_analysiss · Task: Compare three existing tools

My contribution: I checked and found about inferno on HealthIT.gov is a public service for running select FHIR conformance tests built with Inferno and referenced to my team whether us-core tech can be used in our team.

W7-2 · State the gap and your contribution

Logged late · 0 min · 4/4 required parts · 07 Sept, 08:42

The gap in existing solutions

Existing healthcare interoperability tools provide useful capabilities such as synthetic patient-data generation, FHIR server functionality, or automated FHIR conformance testing. However, these solutions do not provide a single, safe and repeatable environment tailored to the MedTalk Connect integration pipeline. In particular, there is a gap in testing the interaction between MedTalk AI, Halo Connect/Best Practice and other clinical systems while also validating the SNOMED CT-AU to ICD-10-AM and MBS coding layer. Existing approaches also do not combine this with controlled testing of invalid data, delayed responses and dropped connections, without using real patient data or production systems.

Your proposed technical contribution to closing it

The proposed contribution is a Synthetic FHIR R4 Sandbox and Test Harness designed specifically to test the backend interoperability requirements of MedTalk Connect. The environment will use synthetic clinical records and FHIR R4 interfaces to simulate interactions with clinical systems such as Best Practice through Halo Connect. It will validate clinical records and SNOMED CT-AU coding, test SNOMED CT-AU to ICD-10-AM and MBS mappings, and introduce controlled fault conditions including malformed data, delayed responses and dropped connections. Automated test execution will record pass/fail outcomes and failure reasons, producing repeatable evidence of whether the integration behaves as expected without accessing real patient or production systems.

Technology or framework you selected

FHIR R4 was selected as the core interoperability framework because the project specifically requires testing structured clinical data exchange through FHIR-compliant interfaces.

One alternative you rejected, and why

Synthea was considered because it can generate realistic synthetic patient records and supports FHIR R4 output. However, Synthea primarily addresses synthetic clinical-data generation rather than the complete interoperability-testing problem. It does not by itself provide the project-specific combination of FHIR integration testing, SNOMED CT-AU to ICD-10-AM/MBS mapping validation, fault injection for degraded network conditions, automated pass/fail evaluation and reporting. Therefore, it was considered useful as a potential data-generation component, but not sufficient as the core technical solution. but it still is considered as reference.

W7-3 · Revise your figure — version 1 to version 2

Logged late · 1 min · 4/4 required parts · 07 Sept, 09:08

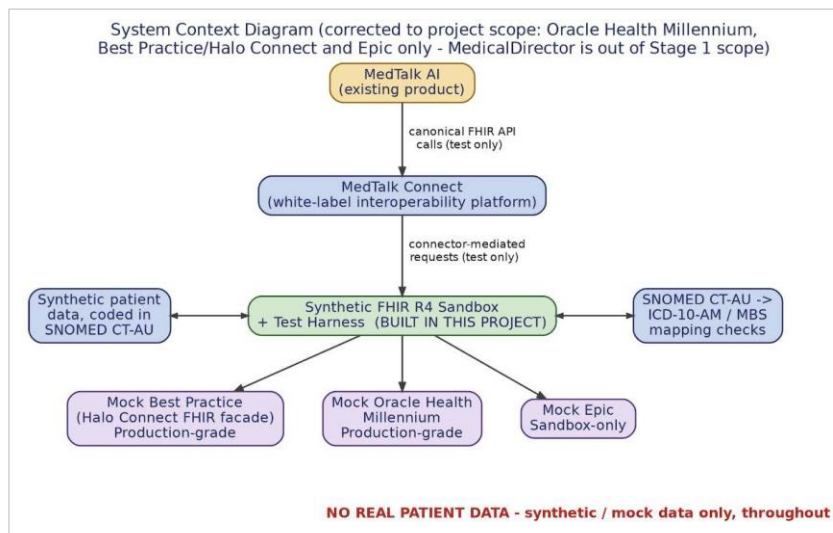
What you changed and why

I worked on the security design, cost estimation, database structure, and mockups for the proposed test harness. I developed the database component to store test data and results, and created mockups to demonstrate how test cases could be executed and identified as passed or failed. I also continued refining the architecture and technology stack from Week 6, while leaving some Week 7 sections still in progress.

Whose feedback changed it

Sonam and charity

W7-3-01 · 07 SEPT 2026



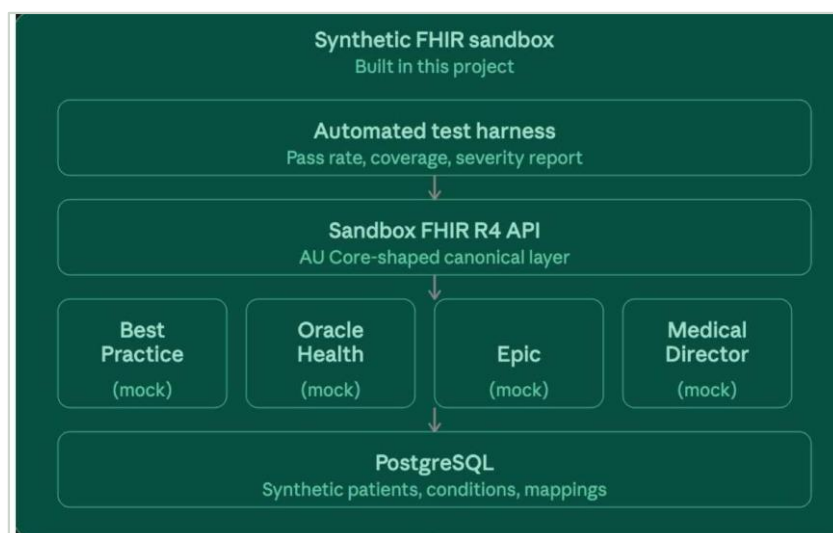
Section: system context · Task: Revise your figure – version 1 to version 2

My contribution: i developed a figure using AI to help sketch a context diagram with details but its just stage 1 and was found not accurate as per my team review no database or 4 mocks shown

What it demonstrates: system context of our test harness

https://github.com/SkinnySeal17/HEALTHCARE_SYSTEM/tree/main/DIAGRAMS

W7-3-02 · 07 SEPT 2026



Section: system context · Task: Revise your figure – version 1 to version 2

My contribution: i sketched it based on version 1 review of our flow diagram including databasae and 4 mocks to test it.

What it demonstrates: overall flow of our test harness

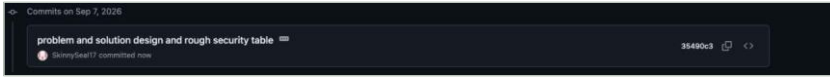
https://github.com/SkinnySeal17/HEALTHCARE_SYSTEM/tree/main/DIAGRAMS

W7-4 · Log a real GitHub action

Logged late · 2 min · 4/4 required parts · 07 Sept, 08:58

DATE	ISSUE / BRANCH / COMMIT REFERENCE	WHAT YOU DID	LINK	OUTCOME
07 sept	riki_updates #3	I developed a figure of solution architecture and got team reviewed and rough security. and cost analysis of our test harness.	https://github.com/SkinnySeal17/HEALTHCARE_SYSTEM/commits/Riki_updates/	already team reviewed and new action to be reviewed.

W7-4-01 · 07 SEPT 2026



Section: solution architecture/ security · Task: Log a real GitHub action

My contribution: I developed a figure of solution architecture and got team reviewed and rough security. and cost analysis of our test harness.

What it demonstrates: Github commit updated after full work completion of section 4 and rough architecture for section 5.

https://github.com/SkinnySeal17/HEALTHCARE_SYSTEM/commits/Riki_updates/

W7-5 · Record one technical decision

Logged late · 0 min · 4/4 required parts · 07 Sept, 08:43

Decision

Adopt FHIR R4 as the standard interoperability boundary for the Synthetic FHIR R4 Sandbox and Test Harness.

Context — what forced the decision

MedTalk Connect must exchange clinical information between MedTalk AI and multiple clinical systems, while the project specifically needs to test this interoperability without accessing real patient or production systems. The existing problem involves validating structured clinical data, coding, responses and failure conditions across system boundaries. Therefore, the solution needs a consistent, standards-based representation of clinical information that can be used to simulate requests and responses between MedTalk and mock clinical systems. FHIR R4 provides this common boundary for the proposed testing environment.

Options considered

Three approaches were considered: FHIR R4, proprietary vendor-specific formats, and a custom data-exchange format. FHIR R4 was selected because it provides a standardised structure for representing and exchanging healthcare information and aligns with the existing MedTalk Connect integration architecture. Vendor-specific formats would make the test harness more dependent on individual systems such as Best Practice, Epic or Oracle Health. A custom format would provide flexibility but would introduce unnecessary complexity and reduce interoperability with the systems being tested.

Consequence, including what gets harder

Using FHIR R4 makes the test environment more consistent and allows interoperability behaviour to be tested against a defined healthcare standard. It also supports clearer validation of requests, responses and clinical resources. However, this decision means the team must correctly implement and validate FHIR R4 structures, resources and required fields. Additional work is also required when simulating vendor-specific behaviour behind the FHIR boundary. The test harness therefore becomes more structured, but that structure comes with implementation and validation overhead.

W7-6 · Weekly reflection

Logged late · 1 min · 4/4 required parts · 07 Sept, 08:49

What did I personally complete this week?

This week, I finalised the rough Week 6 tasks, including the proposed solution architecture and technology stack justification. I also started work on the Week 7 tasks, particularly the Security Design and Control Mapping and Capital and Non-Capital Cost Estimation sections. I produced rough initial work for both sections and identified the main security controls, technologies, and cost areas that need to be included. However, these sections are not fully completed by the end of Week 7.

What did I learn about the problem or the technology?

This week, I developed a better understanding of how the proposed architecture and technology stack support the interoperability requirements of the project. I also learned that security controls need to be mapped directly to the identified risks and system requirements rather than being listed generally. Working on the cost estimation also showed me that the solution involves both technical infrastructure costs and non-capital resources, such as development, testing and maintenance effort.

What decision did I influence, and why?

I influenced the decision to structure the proposed solution around FHIR R4 as the main interoperability boundary for the test harness. This was important because the project focuses on backend system interoperability rather than changes to

the clinician-facing MedTalk scribe. Using FHIR R4 provides a consistent standard for exchanging and validating synthetic clinical data between the test harness and simulated clinical systems, while also aligning the solution with the existing MedTalk Connect integration approach.

What will I finish before the next class?

Before the next class, I will complete the remaining Week 7 deliverables and hopefully sketch the wireframe about security specific flow. This includes finalising the Security Design and Control Mapping section and completing the Capital and Non-Capital Cost Estimation. I will refine the rough work already produced, ensure the security controls are linked to the project requirements and risks, and review the cost estimates for consistency with the proposed architecture and technology stack. I will also check that both sections are ready to be incorporated into the final report.

Week 8

7 of 7 entries logged · 8 minutes recorded · 9 evidence items · 100% complete

W8-1 · Show your component running

Logged late · 1 min · 4/4 required parts · 13 Sept, 13:20

What you built or configured

Inside the healthcare_system repository, we initialized and built a production-ready, client-side routing environment using React 18 and Tailwind CSS. The component features an interactive state machine that eliminates heavy database dependencies by dynamically checking for local variables and browser query strings (iss and launch). We fully programmed the Identity and Access Management Gate, which separates clinician single sign-on triggers from administrative developer testing pathways. It natively enforces local verification strings (mt_live_secret123), handles high-contrast input-error rendering loops, and features a floating testing overlay controller that emulates upstream SMART on FHIR launch sequences without requiring external server calls.

How to run or reproduce it

1. Ensure you are working in your isolated feature branch
git checkout feature/auth-flow-sandbox

2. Navigate to the newly initialized frontend directory
cd healthcare_system/frontend

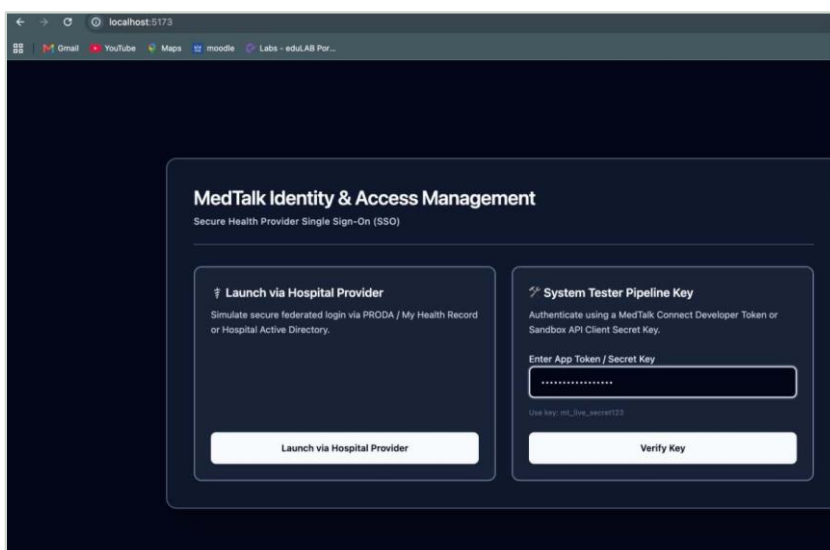
3. Clean install the required Vite, React, and Tailwind dependencies
npm install

4. Spin up the local development web server compile engine
npm run dev

5. Open your browser and navigate to the base standalone manual flow:
<http://localhost:5173/>

6. To reproduce and verify the background SMART on FHIR integration, paste this URL query parameter strings link:
<http://localhost:5173/?iss=https://bestpractice.com.au>

W8-1-01 · 13 SEPT 2026

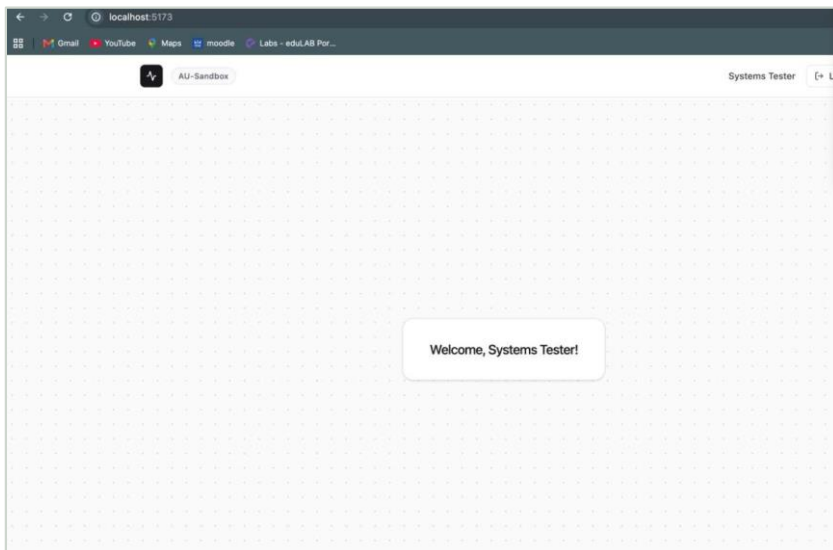


Section: login flow - page 2 · Task: Show your component running

My contribution: i developed page 2 of login flow - it checks whether you are doctor or tester and allows entry to dashboard accordingly.

What it demonstrates: Clicking the button in page 1 flips a switch that brings up the login portal. Instead of typing text passwords, users choose between a quick hospital login button for doctors or a secret token box for testers. Typing the wrong token triggers a red error warning instantly.

W8-1-02 · 13 SEPT 2026

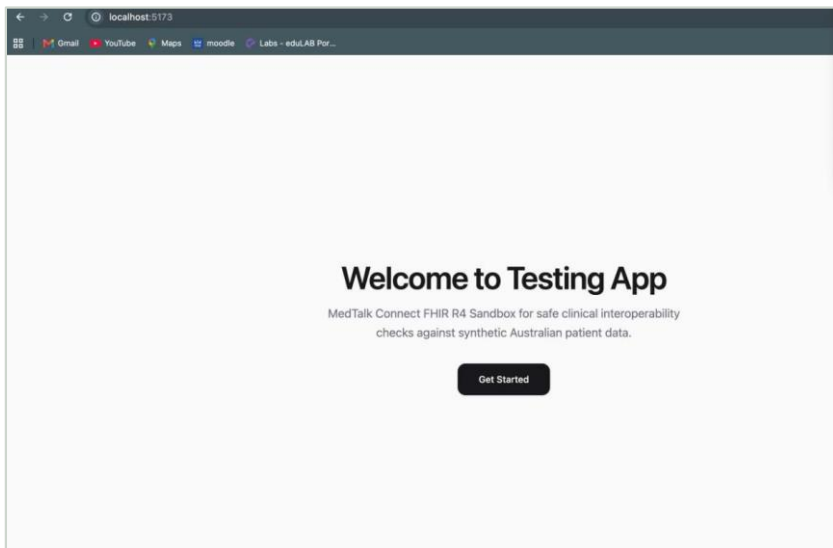


Section: page 3 -login flow · Task: Show your component running

My contribution: i built a home page with "Welcome" banner at the top, ready for the medical tools to be built next

What it demonstrates: Once the login is successful, the app remembers your name and your role (Doctor or Tester). It clears away the login box and loads an empty dashboard layout containing a custom "Welcome" banner at the top, ready for the medical tools to be built next.

W8-1-03 · 13 SEPT 2026



Section: page 1 - login flow · Task: Show your component running

My contribution: i started the first page of our app.

What it demonstrates: When the app first opens, it looks at the browser address bar to see if a medical system is linking to it. If it finds no special links, it stays on a simple welcome screen with a clean title and a single "Get Started" button

W8-1B · Upload your sitemap, wireframe and high-fidelity design

Logged late · 3 min · 7/7 required parts · 13 Sept, 12:56

Tool used

draw.io/ vs code

Sitemap — the screens or pages in your solution and how they connect

my sitemap shows complete visual architecture map of the application's authentication infrastructure. It models how users move from a standard Welcome Screen through a secure Medical Identity and Access Management Gate. The diagram maps out two distinct pathways: a automated SMART on FHIR Launch profile for doctors syncing from downstream clinical software (like Best Practice or Epic) and a Secure App Token Entry path for systems testers. Finally, it outlines how the frontend routes these profiles into dedicated, role-specific dashboard views and logs back out securely.

Wireframe — which screen you wireframed and what a user can do on it

The first wireframe is the Test Scenario Builder for MedTalk Connect a security feature meant to allow all access to tester while only doctors system to their only and 2nd is, where a tester can select a clinical system, choose a test scenario, select a synthetic patient and condition, review terminology mappings, validate the synthetic data, and either save the scenario as a draft or run the test. after running, it will validate pass/fail result.

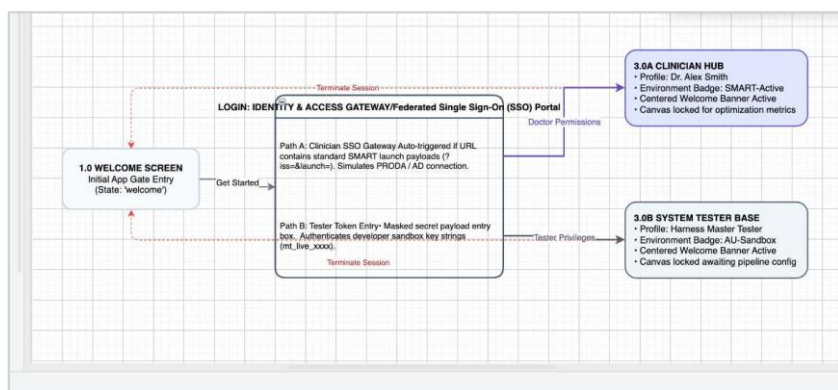
High-fidelity — what changed from the wireframe, and why

(Identity & Access Gateway): A split component container showing professional healthcare integration entry blocks side-by-side: Left Panel: The Clinician EMR Launch Hub with an indicator badge for when an automated background SMART context handshake is detected. Right Panel: The System Tester Key Entry Field, featuring a password-masked developer token block. nothing really changes at this timeframe.

Which parts of these three artefacts did you personally produce?

i personally produced the login flow and smart FHIR flow and also the hi-fi design for the login flow. Also i wireframed the test case scenario page after login flow.

W8-1B-01 · 13 SEPT 2026



Section: sitemsp of login flow · Task: Upload your sitemap, wireframe and high-fidelity design

My contribution: i develop a complete visual architecture map of your application's authentication infrastructure.(initial stage)

What it demonstrates: It models how users move from a standard Welcome Screen through a secure Medical Identity and Access Management Gate. The diagram maps out two distinct pathways: a automated SMART on FHIR Launch profile for doctors syncing from downstream clinical software (like Best Practice or Epic) and a Secure App Token Entry path for systems testers

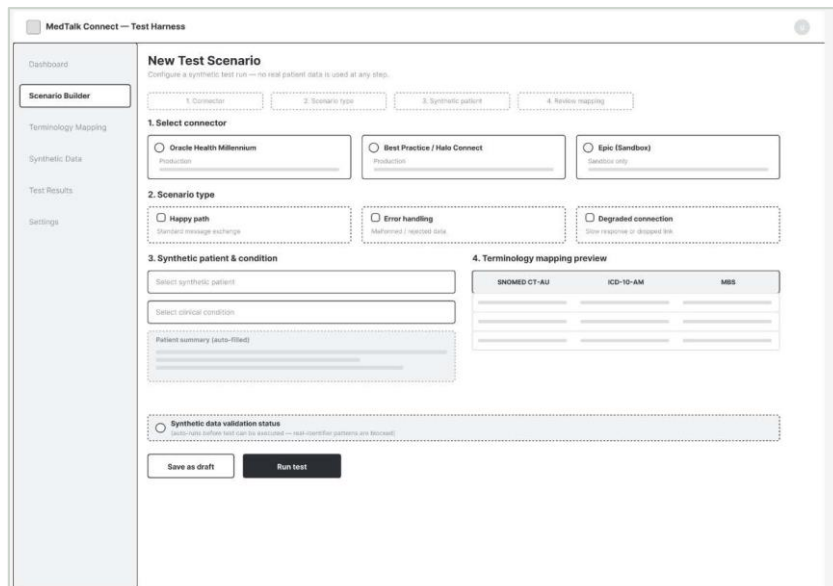
W8-1B-02 · 13 SEPT 2026

Section: hifi design · Task: Upload your sitemap, wireframe and high-fidelity design

My contribution: i develop auth flow of doctors and tester

What it demonstrates: Left Panel: The Clinician EMR Launch Hub with an indicator badge for when an automated background SMART context handshake is detected. Right Panel: The System Tester Key Entry Field, featuring a password-masked developer token block.

W8-1B-03 · 13 SEPT 2026



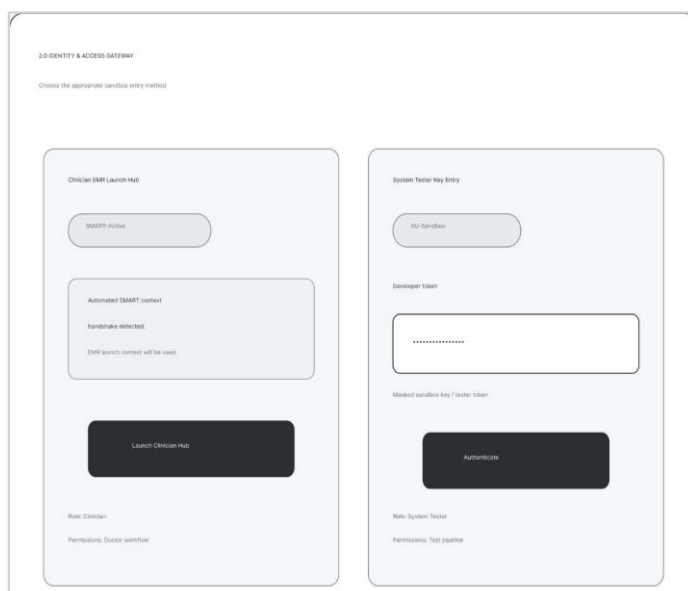
Section: wireframe of dashboard · Task: Upload your sitemap, wireframe and high-fidelity design

My contribution: i developed a wireframe after the login flow of how the dashboard would possibly look

What it demonstrates: THis is the first dashboard diaagram i have drawn for tester after login flow that shows the connector type, scenario and pass/fail system to check the testing of sythetic data

<https://www.figma.com/design/rErJwxv2IimhVMxf4U76gI/Untitled?node-id=0-1&p=f&m=draw>

W8-1B-04 · 13 SEPT 2026



Section: identity and access management login flow wireframe · Task: Upload your sitemap, wireframe and high-fidelity design

My contribution: i developed this access management for doctors and testers that will authenticate them and set them to their respective pages.

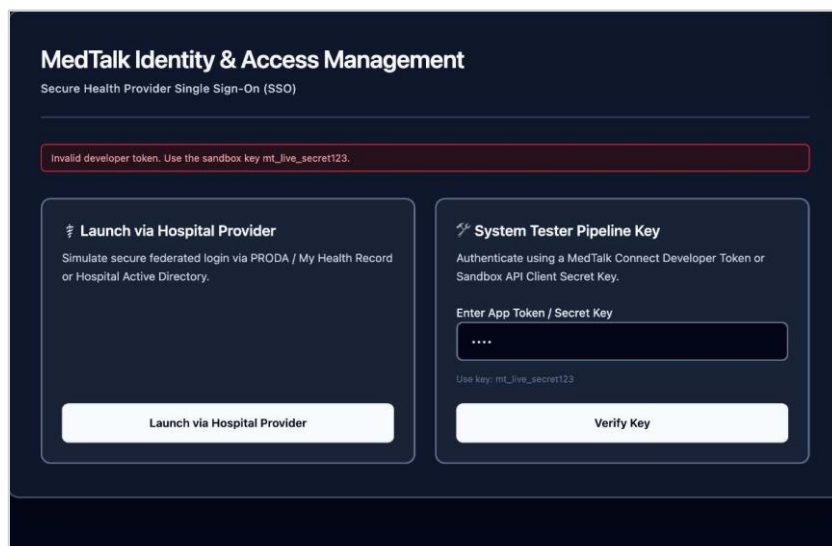
What it demonstrates: a security feature meant to allow all access to tester while only doctors system to their only

W8-2 · Test it and record the results

Logged late · 1 min · 4/4 required parts · 13 Sept, 13:21

TEST ID	WHAT IS BEING TESTED	INPUT / STEPS	EXPECTED	ACTUAL	PASS / FAIL
T-01	Developer Key Submission - Valid Token Access	1. Navigate to Login view.2. In 'System Tester Key' field, input: mt_live_secret123. 3. Select 'Verify Key & Inject Harness'	System halts state progression, preserves the bad string inside the viewport, and exposes an active layout warning alert.	System accepted token string, set userRole state hook to 'Tester', and exposed the layout with the 'AU-Sandbox' environment tag.	pass
T-02	Developer Key Submission - Invalid Token Rejection	Navigate to Login view.2. In 'System Tester Key' field, input: wrong_key_456. 3. Select 'Verify Key & Inject Harness'.	System halts state progression, preserves the bad string inside the viewport, and exposes an active layout warning alert.	State progression blocked. The component rendered a high-contrast red warning block explaining key mismatch criteria. Input borders turned red.	pass
T-03	launch hospital provider login	navigate to login and click launch via hospital provider	System bypasses manual token field entries, forces the user profile state to 'Doctor', and loads the base empty dashboard.	Local state router successfully captured the trigger event, resolved userName to 'Dr. Sarah Jenkins', and rendered the locked clinician layout.	pass

W8-2-01 · 13 SEPT 2026



Section: login fail for tester · Task: Test it and record the results

My contribution: i update your frontend to validate the key locally based on simple string matching rules.

What it demonstrates: I didnt use the correct string so i cannot login and the system displays use corerct key and i added the correct token key there for testing purpose only which need to be removed during fututre deployment.

W8-3 · Map your work to a security framework

Logged late · 1 min · 3/3 required parts · 13 Sept, 13:12

Framework used

NIST Cybersecurity Framework 2.0

CONTROL REFERENCE	CONTROL NAME	HOW YOUR COMPONENT ADDRESS	FULL / PARTIAL / PLANNED
PR.AA-01	Identity and credential management	The PoC provides separate clinician and system-tester entry paths, with authentication through the SSO/SMART launch flow or a masked tester token.	partial
PR.AA-05	Access permissions and authorisations	The design separates Clinician and System Tester roles and shows different permissions for each role. Actual backend enforcement is not demonstrated in the wireframe.	partial
PR.DS-02	Data-in-transit protection	The sandbox is designed for secure healthcare-system integration and authentication, but the PoC does not demonstrate actual TLS/encryption implementation.	partial

W8-4 · Ethics, privacy, legal and AI governance

Logged late · 1 min · 3/3 required parts · 13 Sept, 13:22

Personal data your solution touches, and the obligation that applies

Our sandbox processes synthetic patient identifiers, clinician profile mappings, and mock endpoint keys. Any accidental input of real staff data invokes the Privacy Act 1988 (Cth). Under APP 11, we must actively protect this information from misuse or exposure. We fulfill this by isolating all login states within local, short-lived browser memory (useState), completely avoiding unencrypted server storage.

AI use in this project and how it was governed

We utilized Cursor AI to generate React templates, construct the drawing schematics, and write Tailwind CSS layouts. Governance was strictly enforced via a root .cursorrules file. We manually verified the query string interceptors to prevent malformed code bugs, and rejected AI suggestions to use insecure local cookies or generic password forms that violate health standards.

One ethical or legal risk you cannot fully mitigate

We cannot fully mitigate "Configuration Drift to Production." Testers could accidentally copy sandbox configurations or raw API key structures from this branch into a live hospital network. While our UI prominently flags the "AU-Sandbox" environment, we cannot technically block external copy-paste human errors, leaving an open legal risk for external deployment gatekeepers.

W8-5 · Own a slide

Logged late · 0 min · 5/5 required parts · 13 Sept, 13:45

Slide title and number you are responsible for

Login flow and solution design

The three points you will make

Login flow separates clinician SSO from tester token access — two trust boundaries from screen one. Security controls map cleanly to NIST CSF, anchored at the login gateway.- identify, protect, detect, respond/ recover. Stack and ~\$0 AUD cost are justified by interoperability fit and the project's zero-budget constraint, not convenience.

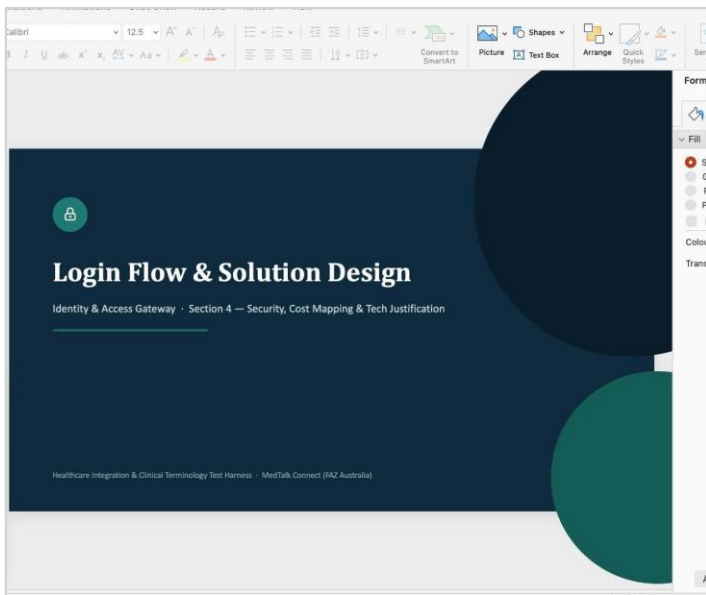
A question you expect from the audience, and your answer

why we need auth in synthetic data?

Even synthetic data and test configurations need access control. Without it, anyone could alter test scenarios, corrupt terminology mappings, or mark results as validated when they aren't — undermining the integrity of the harness before it's ever trusted with real connectors in Stage 2. Authentication here isn't protecting patient data directly; it's protecting the

trustworthiness of the test evidence itself.

W8-5-01 · 13 SEPT 2026



Section: presentation slide · Task: Own a slide

My contribution: i developed 6 slides for my section of login flow and solution design i created

What it demonstrates: the first page of my slides

W8-6 · Limitations, peer cross-check and final reflection

Logged late · 0 min · 7/7 required parts · 13 Sept, 13:28

Honest limitations of your contribution

The primary limitation is that all routing, authentication, and error validations are currently mocked on the client side using local React state. There is no live Python backend or SQL database attached yet to persist logins or verify tokens over a network. Additionally, the SMART on FHIR launch only emulates query string parsing (?iss= and ?launch=) inside the browser. It does not perform actual cryptographic validation or exchange real JSON Web Tokens (JWT) with an official identity provider.

Which group member's evidence did you review today?

sonam

One specific thing that was strong and one thing that needs work in their evidence

Strong: The RACI matrix clearly assigns ownership by domain (e.g. Charity for clinical data, Rikesh for technical integration), which made accountability easy to follow.

Needs work: The Gantt chart / roadmap section is still a placeholder — it needs actual dates and milestones added before Stage 2 planning can be checked against it.

What did I personally complete this week?

I initialized the healthcare_system frontend repository branch using Vite and React from scratch. I built the single-file state router implementing the Welcome Gate, the Split-Role Authentication Selector, and the Base Dashboard Layout. Furthermore, I programmed local validation handling to reject bad developer keys, integrated standard SMART on FHIR query interceptors, and generated matching architecture sitemaps and wireframe layout sheets for draw.io.

What did I learn about the problem or the technology?

I learned how critical federated Single Sign-On (SSO) and launch parameters are in medical tech compared to standard apps. Doctors face immense digital fatigue; typing standard passwords for every logging tool adds cognitive load. Interoperability protocols like SMART on FHIR solve this by safely passing session states directly through the EHR interface, proving that authentication design is deeply tied to healthcare workflow optimization.

What decision did I influence, and why?

I influenced the group decision to completely bypass traditional local password databases in our repository. I argued that

building standard user signup forms is a security vulnerability and does not reflect real Australian digital health infrastructure. Pivoting to a federated, token-based mock gateway better demonstrates high-level engineering compliance with modern healthcare software architectures.

What will I finish before the next class?

I completed the login flow (authentication screen) as planned. Before the next class, I'll finish preparing my section for the presentation and be ready to take feedback on it. Its still in initial stage, ill ask feedback from the mentor and see how he wants the flow.

Literature and evidence tracker

WK	SOURCE	SEARCH METHOD / ORIGIN	KEY INFORMATION COLLECTED	HOW IT WAS USED
7	Synthea (MITRE Corporation / Open source, Apache 2.0)	Github	Generates realistic but synthetic patient populations and clinical histories. Supports FHIR R4, including conditions, medications, observations, procedures and other clinical resources. Useful for producing safe test data without real patient information.	Primarily focuses on synthetic patient/data generation. It does not by itself provide the complete integration-testing workflow required for FHIR exchange, fault injection, S N O M E D C T - A U I C D - 1 0 mapping validation and automated test reporting.
7	HAPI FHIR (Smile CDR / HAPI FHIR open-source project)	https://hapifhir.github.io/hapi-hl7v2/license.html	Provides a FHIR implementation and server framework that can be used to store and exchange FHIR resources. Useful for creating a FHIR-based test environment and testing API interactions.	Provides the FHIR infrastructure, but the project still needs additional testing logic for the specific requirements of this project, such as fault injection, mapping validation, automated pass/fail evaluation and project-specific reporting.
7	Inferno (MITRE / ONC ecosystem, open source, Apache 2.0)	https://fhir.healthit.gov/test-kits/	Automated FHIR conformance testing framework. Provides executable test kits for FHIR implementation guides and can test FHIR APIs against defined requirements. It also provides simulated FHIR APIs for some test kits.	Strongly focused on FHIR conformance testing and specific implementation guides. It does not directly provide the complete combination of synthetic clinical data generation, SNOMED C T - A U ! I C D - 1 0 mapping checks, healthcare-specific fault scenarios and the customised test harness required by this project.
8	NIST Cybersecurity Framework 2.0 PR.AA-01	framework document	Identity and credential management	The PoC provides separate clinician and system-tester entry paths, with authentication through the SSO/SMART launch flow or a masked tester token.
8	NIST Cybersecurity Framework 2.0 PR.AA-05	framework document	Access permissions and authorisations	The design separates Clinician and System Tester roles and shows different permissions for each role. Actual backend enforcement is not demonstrated in the wireframe.
8	NIST Cybersecurity Framework 2.0 PR.DS-02	framework document	Data-in-transit protection	The sandbox is designed for secure healthcare-system integration and authentication, but the PoC does not demonstrate actual TLS/encryption implementation.

GitHub and project-management activity

DATE	PLATFORM	REF	ACTION	LINK	OUTCOME
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DATE	PLATFORM	REF	ACTION	LINK	OUTCOME
07 sept	GitHub	riki_updates #3	I developed a figure of solution architecture and got team reviewed and rough security. and cost analysis of our test harness.	https://github.com/Skianready team nySeal17/HEALTHCARE_SYSTEM/commits /Riki_updates/	reviewed and new action to be reviewed.

Evidence index

NO.	COLLECTED	SECTION	TASK	WHAT IT DEMONSTRATES
S-2-01	07 Sept 2026	—	Responsibility split	
W7-1-01	07 Sept 2026	AS_is _analysis	Compare three existing tools	ONC Certification (g)(10) Standardized API Test Kit and US Quality Core Test Kit research
W7-3-01	07 Sept 2026	system context	Revise your figure — version 1 to version 2	system context of our test harness
W7-3-02	07 Sept 2026	system context	Revise your figure — version 1 to version 2	overall flow of our test harness
W7-4-01	07 Sept 2026	solution architecture/ security	Log a real GitHub action	Github commit updated after full work completion of section 4 and rough architecture for section 5.
W8-1-01	13 Sept 2026	login flow - page 2	Show your component running	Clicking the button in page 1 flips a switch that brings up the login portal. Instead of typing text passwords
W8-1-02	13 Sept 2026	page 3 -login flow	Show your component running	Once the login is successful, the app remembers your name and your role (Doctor or Tester). It clears away the
W8-1-03	13 Sept 2026	page 1 - login flow	Show your component running	When the app first opens, it looks at the browser address bar to see if a medical system is linking to it. If
W8-1B-01	13 Sept 2026	sitemap of login flow	Upload your sitemap, wireframe and high-fidelity design	It models how users move from a standard Welcome Screen through a secure Medical Identity and Access Management
W8-1B-02	13 Sept 2026	hifi design	Upload your sitemap, wireframe and high-fidelity design	Left Panel: The Clinician EMR Launch Hub with an indicator badge for when an automated background SMART context
W8-1B-03	13 Sept 2026	wireframe of dashboard	Upload your sitemap, wireframe and high-fidelity design	This is the first dashboard diagram I have drawn for tester after login flow that shows the connector type, s
W8-1B-04	13 Sept 2026	identity and access management login flow wireframe	Upload your sitemap, wireframe and high-fidelity design	a security feature meant to allow all access to tester while only doctors system to their only
W8-2-01	13 Sept 2026	login fail for tester	Test it and record the results	I didn't use the correct string so I cannot login and the system displays use correct key and I added the correct
W8-5-01	13 Sept 2026	presentation slide	Own a slide	the first page of my slides

VERIFICATION

Time log

Each entry below was recorded automatically while the task was open in the portfolio tool. The timer pauses after three minutes without activity.

TASK	SESSION START	DURATION	CONTEXT
S-2	24 Aug, 10:30	00:11	In class
W6-1	24 Aug, 10:30	00:13	Outside class
W6-2	24 Aug, 10:31	00:05	Outside class
S-2	24 Aug, 10:37	00:15	In class
W6-1	24 Aug, 10:37	00:29	Outside class
W6-1	24 Aug, 10:47	00:27	Outside class
W6-1	31 Aug, 09:58	00:15	In class
W7-1	31 Aug, 09:59	00:21	In class
W7-1	07 Sept, 08:26	00:22	Outside class
W7-2	07 Sept, 08:26	00:12	Outside class
W7-5	07 Sept, 08:26	00:11	Outside class
W7-1	07 Sept, 08:27	00:07	Outside class
W7-2	07 Sept, 08:38	00:16	Outside class
W7-5	07 Sept, 08:42	00:09	Outside class
W7-6	07 Sept, 08:43	00:07	Outside class
W7-6	07 Sept, 08:47	00:30	Outside class
W7-6	07 Sept, 08:47	00:10	Outside class
W7-4	07 Sept, 08:55	00:30	Outside class
W7-4	07 Sept, 08:55	00:31	Outside class
W7-4	07 Sept, 08:56	00:31	Outside class
W7-4	07 Sept, 08:56	00:23	Outside class
W7-3	07 Sept, 08:58	00:18	Outside class
S-2	07 Sept, 09:13	00:30	Outside class
S-2	07 Sept, 09:13	00:15	Outside class
W8-1B	07 Sept, 09:29	00:16	In class
W7-3	07 Sept, 10:47	00:22	Outside class
W8-1	11 Sept, 22:52	00:29	Outside class
W8-1B	11 Sept, 22:52	00:19	Outside class
W8-2	11 Sept, 23:13	00:30	Outside class
W8-3	11 Sept, 23:13	00:20	Outside class
W8-4	11 Sept, 23:13	00:30	Outside class
W8-4	11 Sept, 23:14	00:30	Outside class
W8-4	11 Sept, 23:14	00:11	Outside class

TASK	SESSION START	DURATION	CONTEXT
W8-1B	11 Sept, 23:15	00:27	Outside class
W8-1B	13 Sept, 11:53	00:08	Outside class
W8-1	13 Sept, 12:33	00:30	Outside class
W8-1	13 Sept, 12:34	00:05	Outside class
W8-3	13 Sept, 12:35	00:14	Outside class
W8-1B	13 Sept, 12:36	00:29	Outside class
W8-1B	13 Sept, 12:51	00:30	Outside class
W8-1B	13 Sept, 12:52	00:30	Outside class
W8-1B	13 Sept, 12:52	00:16	Outside class
W8-2	13 Sept, 13:03	00:30	Outside class
W8-3	13 Sept, 13:11	00:07	Outside class
W8-1	13 Sept, 13:12	00:10	Outside class
W8-2	13 Sept, 13:20	00:16	Outside class
W8-4	13 Sept, 13:21	00:09	Outside class
W8-6	13 Sept, 13:22	00:12	Outside class
W8-5	13 Sept, 13:28	00:14	Outside class

Weekly individual reflection

Week 7

What did I personally complete this week?

This week, I finalised the rough Week 6 tasks, including the proposed solution architecture and technology stack justification. I also started work on the Week 7 tasks, particularly the Security Design and Control Mapping and Capital and Non-Capital Cost Estimation sections. I produced rough initial work for both sections and identified the main security controls, technologies, and cost areas that need to be included. However, these sections are not fully completed by the end of Week 7.

What did I learn about the problem or the technology?

This week, I developed a better understanding of how the proposed architecture and technology stack support the interoperability requirements of the project. I also learned that security controls need to be mapped directly to the identified risks and system requirements rather than being listed generally. Working on the cost estimation also showed me that the solution involves both technical infrastructure costs and non-capital resources, such as development, testing and maintenance effort.

What decision did I influence, and why?

I influenced the decision to structure the proposed solution around FHIR R4 as the main interoperability boundary for the test harness. This was important because the project focuses on backend system interoperability rather than changes to the clinician-facing MedTalk scribe. Using FHIR R4 provides a consistent standard for exchanging and validating synthetic clinical data between the test harness and simulated clinical systems, while also aligning the solution with the existing MedTalk Connect integration approach.

What will I finish before the next class?

Before the next class, I will complete the remaining Week 7 deliverables and hopefully sketch the wireframe about security specific flow. This includes finalising the Security Design and Control Mapping section and completing the Capital and Non-Capital Cost Estimation. I will refine the rough work already produced, ensure the security controls are linked to the project requirements and risks, and review the cost estimates for consistency with the proposed architecture and technology stack. I will also check that both sections are ready to be incorporated into the final report.

Week 8

What did I personally complete this week?

I initialized the healthcare_system frontend repository branch using Vite and React from scratch. I built the single-file state router implementing the Welcome Gate, the Split-Role Authentication Selector, and the Base Dashboard Layout. Furthermore, I programmed local validation handling to reject bad developer keys, integrated standard SMART on FHIR query interceptors, and generated matching architecture sitemaps and wireframe layout sheets for draw.io.

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I influenced the group decision to completely bypass traditional local password databases in our repository. I argued that building standard user signup forms is a security vulnerability and does not reflect real Australian digital health infrastructure. Pivoting to a federated, token-based mock gateway better demonstrates high-level engineering compliance with modern healthcare software architectures.

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I completed the login flow (authentication screen) as planned. Before the next class, I'll finish preparing my section for the presentation and be ready to take feedback on it. It's still in initial stage, I'll ask feedback from the mentor and see how he wants the flow.

Declaration

I confirm that the work recorded in this portfolio is my own individual contribution to the group project. Screenshots and links refer to work I personally performed. Where AI tools were used, this is disclosed in the Week 8 governance entry.

Signed: _____ Date: _____